

Save, then shop

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Hot water

By reducing the temperature setting of your water heater from 60 to 50 degrees Celsius, you could save over 18 per cent of the energy used at the higher setting

Go green = Save money

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Usually when we think of 'going green', we assume it's for people with deep pockets – that it will involve making monetary sacrifices and taking a lot of trouble, going out of the way in terms of both effort and money. This is just a widespread misconception, because on the contrary, cutting corners by taking green measures will save you a lot of money. Here, we tell you eight simple steps you can take that will help you save money and even save the planet.

1 Switch to CFLs or LEDs

Environmentalists have been exhorting this paradigm shift for quite some time now. If you still have incandescent bulbs then you seriously need to throw them out. Consider this: by some estimates, the life span of LED bulbs is around 45,000 to 60,000 hours. On the other hand, incandescent bulbs and CFLs operate for 1,500 and 10,000 hours, respectively. In terms of luminescence or the amount of light each of these produce, to get the equivalent of 60 W of light from an incandescent bulb, you only have to consume around 14 W for a CFL and 6 W for an LED. You might then say that the initial investment of the environmentally friendly options would be high. Yes, but considering the life spans and the low power consumption of both CFL and LED, you'd recover the cost in no time. In fact, imagine the money saved over the entire life span of an LED bulb – 60,000 hours. By some estimates, over its life span, it'll save you close to Rs. 16,000 in cost of bulbs (40 incandescents)

as well as electricity cost. But a figure you can relate to more is how much the switch from incandescent bulbs will save you in simple energy costs for a year - Rs. 788 for two bulbs.

2 Driving tips

It's a commonly accepted fact that driving between 45 to 55 kmph increases fuel efficiency. At this optimal speed bracket, you're expected to drive steadily without jerky acceleration and rapid braking. Basically, don't be an aggressive driver and it'll save you 15 per cent of your fuel cost; that's the thumb rule. Idling the car for a long time obviously makes no sense. You're wasting fuel and burning money, adding to the pollution, and not going anywhere. The rule here is that if you're going to stop for more than 30 seconds, you should switch off the engine. The initial surge of fuel burn at ignition gets countered if the wait time is above 30 seconds. A Japanese study showed that for a distance of 3,700 kms, when idling norms are strictly followed it saves 6 per cent. For an average Indian car with a mileage of 15 kmpl this distance would cost around Rs. 12,000 of fuel. So you could potentially save Rs. 740, every 3,700 kilometres. Consider this: if you travel 30 km a day, just preventing your car from idling can save you 6 per cent of $30 \times 365 \times 50 / 15 = \text{Rs. } 2,190$ a year!

Also remember to use properly inflated tyres, avoid unnecessary dependence on air conditioning, and whenever possible, carpool.

Eight simple steps that can help you save enough to subscribe to Digit for 9 years!

3 Switch to solar heating

One of the biggest guzzlers of electricity in any household is the big old water heater. In some parts of the country water is heated for nearly half of the year. This water is not used just for bathing, but also for other applications – washing hands, cooking, etc. A typical storage water heater that can raise the temperature of 25 litres by 35 degrees Celsius has a rating of 2000 W and takes about 33 minutes to do it. So a reasonable estimate would be that you'd need an average of 2 cycles to heat about 100 litres of water, ie one hour of usage a day (considering some water will not have to be raised all the 35 degrees from room temperature). This means 2000 watts a day $\times 30 = 60$ kwh. Comes out to Rs. 240 per month just on heating water. By way of instant heaters the amount is almost the same. At a flow of 3 litres per minutes most instant heaters raise the temperature by 21 degrees. So heating around 100 litres of water takes about half an hour of runtime. However instant heaters have a higher rating than storage water heaters – 4500 W. That means 15 hours a month $\times 4500 \text{ W} = 67.5$ kWh. At a cost of Rs. 4 per unit that comes out to Rs. 270.

By either of these methods you end up spending over Rs. 1,500 a year on heating water if you only consume hot water for half the year. Switching to solar is a viable cost effective alternative. They work in any climate and the only fuel they need is sunlight – which is free. There is of course an investment which you can quite easily

Yearly household energy savings

	Incandescent Light Bulbs	CFL	LED
Consumption in Watts	60	14	6
KWh used per year @ 5 hrs/day (2 bulbs)	219	51.1	21.9
Electricity Cost (@ Rs 4.00)	Rs. 876	Rs. 204	Rs. 87
Savings by switching from Incandescent	0	Rs. 671	Rs. 788

HOW TO CALCULATE ELECTRICITY CONSUMPTION

Every appliance has a power rating in Watts. This is known as wattage. Multiply this figure with the number of hours you are expected to use that appliance. Divide this to get the Kilo-watt per hour consumption. You can use this formula:

$\text{Wattage} \times \text{Hours Used Per Day} \div 1000 = \text{Daily Kilowatt-hour (kWh) consumption}$

A Kilowatt-hour is also called a unit. So if an appliance consumes 3kWh it uses 3 units. Multiply this by the prevalent rate per unit and you get the energy consumption in Rupees.